

NGUYEN GIA HUY

FinTech Backend Engineer – Core Banking

johnnynnguyen882@gmail.com | 0948385149 | linkedin.com/in/huy8bit

PROFESSIONAL SUMMARY

Core banking engineer at Vikki Digital Bank, building the **Lending** and **Overdraft** products on **Thought Machine Vault Core** (interest accrual, penalty and late fees, repayment, and settlement) together with the integrations, account migration system, test infrastructure, and AWS deployment around them. Nearly three years of experience, including fintech cloud security at VNPAY (multi-tenant authentication for **20+** fintech and enterprise customer organizations, gateway access control for **100+** services) and transaction integrity across **5+** blockchain networks.

TECHNICAL SKILLS

Core Banking:	Thought Machine Vault Core, Python Smart Contracts, product simulation testing
Banking Products:	Lending, overdraft; interest accrual, penalty and late fees, repayment, settlement
Languages:	Python, Golang, SQL
Integration & Data:	gRPC, REST/JSON, Kafka, PostgreSQL, Amazon RDS, Redis
Security:	Keycloak (multi-tenant RBAC), Kong Gateway, HashiCorp Vault
Cloud:	AWS (EKS, ECS, Lambda, RDS, ECR, ELB), Kubernetes
DeFi:	Solidity, Move, Ethereum, BSC, Polygon, Aptos, Sui

WORK EXPERIENCE

Middle Software Engineer, Core Banking 03/2026 – Present

Galaxy FinX (Vikki Digital Bank) | *core banking platform built on Thought Machine Vault Core*

Lending and Overdraft Products

- Developed and maintained the bank's **Lending** and **Overdraft** products as Python Smart Contracts on **Thought Machine Vault Core**.
- Implemented **interest accrual** for both products, scheduled through Vault Core events.
- Implemented **penalty and late-fee calculations** for overdue obligations.
- Implemented **repayment** processing, applying customer payments against what is owed.
- Implemented **settlement** logic, the path that moves money and closes out customer obligations.

Banking Integration and Account Migration

- Integrated Vault Core with **external backend services** to support core banking operations and transaction workflows across the bank's systems.
- Built the **account migration system** in Golang on AWS ECS, Lambda, and RDS.
- Cut migration time from **2.5 hours to 27 minutes (5x faster)**, shrinking the time window each migration needs.

Verification and Delivery

- Rebuilt the product **simulation testing framework** in Golang, replacing the Gauge-based framework used to verify contract behaviour.
- Reduced simulation runs from **24 minutes to just over 50 seconds (20x faster)**.
- Deployed a new Vault Core version on AWS, provisioning and integrating **EKS, RDS, ECR, and ELB**.
- Built the **CI/CD pipeline** for automated build, test, and deployment of the core banking platform.

Software Engineer 04/2025 – 03/2026

Vietlink (KDDI Agile Development Center Group) | *au Denki – large-scale electric utility platform serving millions of users*

- Designed, built, and operated ETL pipelines on AWS (Lambda, ECS, RDS, S3, Athena) for large-scale utility data used in analytics and operations.
- Increased batch throughput by **150%** through pipeline redesign, parallel execution, and more efficient data access.
- Migrated search and analytics from **Elasticsearch to OpenSearch**, redesigning ingestion and indexing for stability, consistency, and query performance.
- Built the platform's infrastructure with Terraform and AWS SAM and its CloudWatch and DataDog monitoring for health, failures, and anomalies.

Software Engineer – Fintech Cloud Platform

04/2024 – 04/2025

VNPAY | *VNPAY Cloud – cloud platform for fintech and enterprise customers*

- Implemented multi-tenant **authentication** with **Keycloak** for **20+ fintech and enterprise customer organizations** on VNPAY Cloud.
- Implemented role-based **authorization** that enforces tenant isolation, so each customer organization's access stays separate.
- Enforced service-level and network-level access control for **100+ microservices** through **Kong Gateway**.
- Operated **50+ production Kubernetes clusters** on OpenStack for fintech and enterprise customers.
- Automated provisioning, scaling, and zero-downtime upgrades of those clusters with Cluster API, keeping customer workloads available during upgrades.
- Automated about **80%** of data center provisioning and infrastructure configuration.

Blockchain Developer – DeFi & Transaction Systems

06/2023 – 09/2023

EsoiLabs | *Innovaz Market Factory – cross-chain NFT and token platform*

- Developed and reviewed **15+ smart contracts** for DeFi and NFT use cases in Solidity and Move.
- Integrated backend services with **5+ blockchain networks** (Ethereum, BSC, Polygon, Aptos, Sui).
- Ensured **consistent transaction handling** across chains.
- Preserved **data integrity** between on-chain events and off-chain records.
- Built event-driven services that capture and synchronize on-chain events into off-chain databases and APIs.

SELECTED PROJECTS

DattingQ | Software Engineer

Dating platform

- Built a gRPC-Gateway, Kafka, MongoDB, and Redis backend serving up to **1.5K requests/second** at **80–100 ms**.
- Implemented multi-cluster failover and observability (OpenTelemetry, Prometheus, Grafana, ELK, ClickHouse) for **99.95% uptime**.

MeetQ | Software Engineer

AI-powered online meeting platform

- Built a real-time meeting platform on LiveKit with OpenAI models for live transcription, voice translation, and summarization, and designed its real-time translation pipeline.

EDUCATION & CERTIFICATIONS

University of Science, Ho Chi Minh City (HCMUS)

B.Sc. Information Technology – Knowledge Engineering

DevOps on AWS Specialization | Coursera